

Taylor Kirk

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SKILLS

Generative AI & LLMs : LangChain, LangGraph, RAG, Fine-tuning, Prompt Engineering, OpenAI API, Anthropic
Data Analysis & ML : Python, R, SQL, Pandas, NumPy, Scikit-learn, TensorFlow, Matplotlib, Seaborn
Infrastructure & Tools : Postgres, Redis, MLFlow, Hugging Face, Azure, AWS Sagemaker, Streamlit, Git

EXPERIENCE

LiaiZen

Mar. 2025 – Oct. 2025

AI Engineering Intern

San Diego, CA

- Architected a multi-node **LangGraph** workflow for real-time communication moderation, utilizing **Redis** for stateful session management and **Postgres** for persistent audit logging.
- Integrated **LangChain** guardrails to detect and reshape toxic communication while maintaining conversational tone.

Upwork (Freelance Contract)

Jan. 2025 – Sept. 2025

AI/ML Engineer

Remote

- Engineered a therapy-aid application using **LLMs** to assist students through role-play of specific therapeutic techniques.
- Developed **ML pipelines** with **scikit-learn** and **MLflow** to classify and evaluate patient-therapist interactions.
- Designed a **synthetic data generation** workflow to fine-tune models for realistic and diverse patient profiles.

Aya Healthcare

Oct. 2014 – Sept. 2024

Senior Recruiter

San Diego, CA

- Placed **1,500+ travel clinicians**, contributing to organizational growth from 600 to 40,000+ employees.
- Recognized with **Circle of Excellence (4x)** for top-tier performance and process optimization.

TECHNICAL PROJECTS

Offline RL for Clinical Decision Support | Python, PyTorch, LSTM, CQL, MIMIC-IV | Live App

- Built a two-stage pipeline on ~396K ED stays: an **LSTM** sequence model (F1 0.93, PR-AUC 0.98) generating transfer-likelihood estimates at each clinical event to construct a downstream reward function.
- Trained a **Conservative Q-Learning (CQL)** offline RL agent to recommend care escalation; policy achieved **F1 macro 0.85** and flagged ICU transfer a median of **4 clinical events earlier** than the treating provider.

Sentiment Analysis for Political Discourse | Python, Transformers, BERTweet, Hugging Face

- Fine-tuned **BERTweet-large** on political social media comments to classify discourse function across a custom 5-label schema (Argumentative, Informational, Opinion, Expressive, Neutral), achieving **78.17% f1_macro**.
- Published the labeled dataset and trained model to **Hugging Face**, enabling reproducibility and downstream research.

Global Warming Trajectories | R, ARIMA/SARIMA, Shiny | Live App

- Modeled the relationship between anthropogenic CO₂ emissions and global temperature change using **SARIMA** with lagged CO₂ and **ENSO index** as external regressors, achieving **RMSE = 0.1356**.
- Built an interactive **R Shiny** application enabling users to test emission scenarios and visualize projected temperature impacts.

EDUCATION

University of San Diego

May 2024 – Apr. 2026

M.S. Applied Data Science

San Diego, CA

- **Coursework**: Applied Predictive Modeling, Time Series Analysis, Machine Learning, Applied LLMs.

CERTIFICATIONS

AI Essentials | Ready Tensor (RAG Expert)